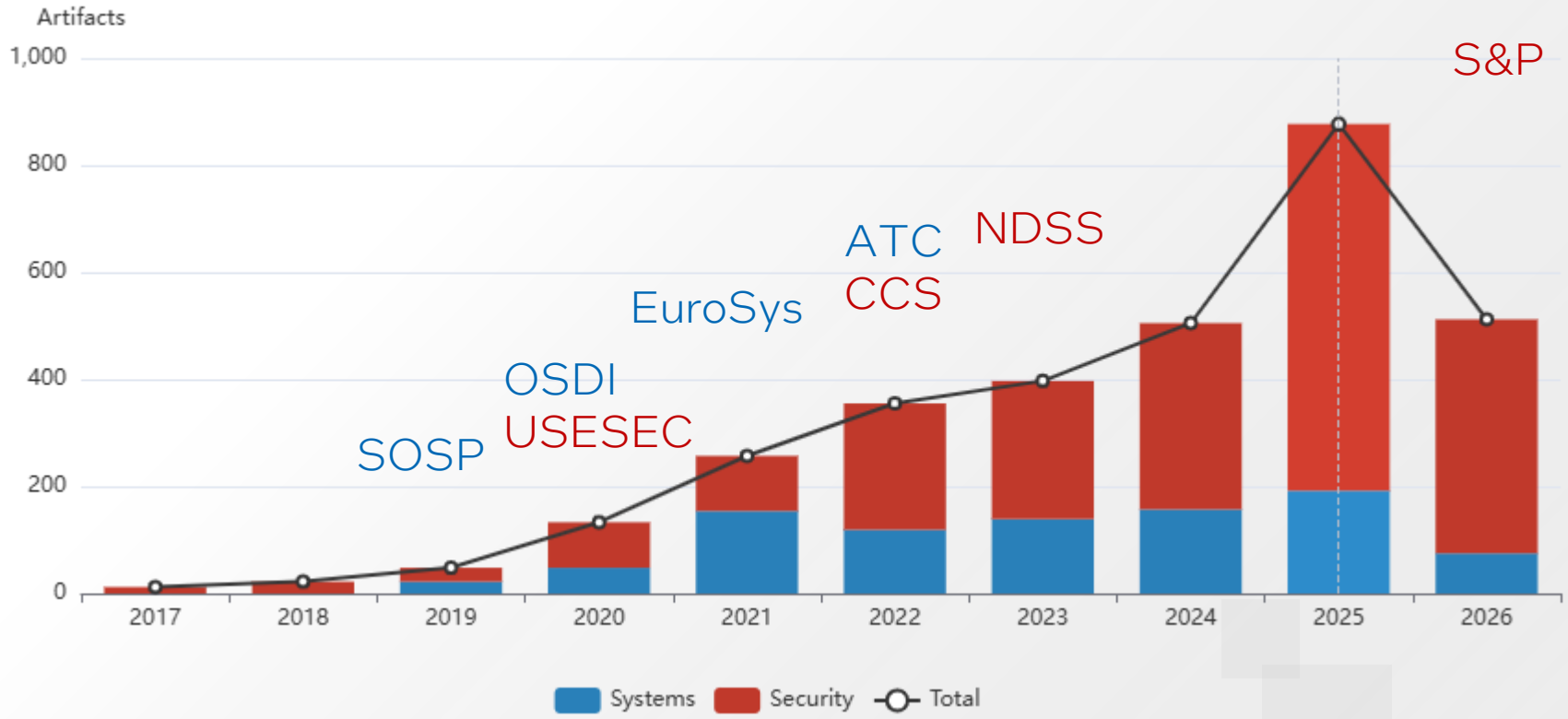
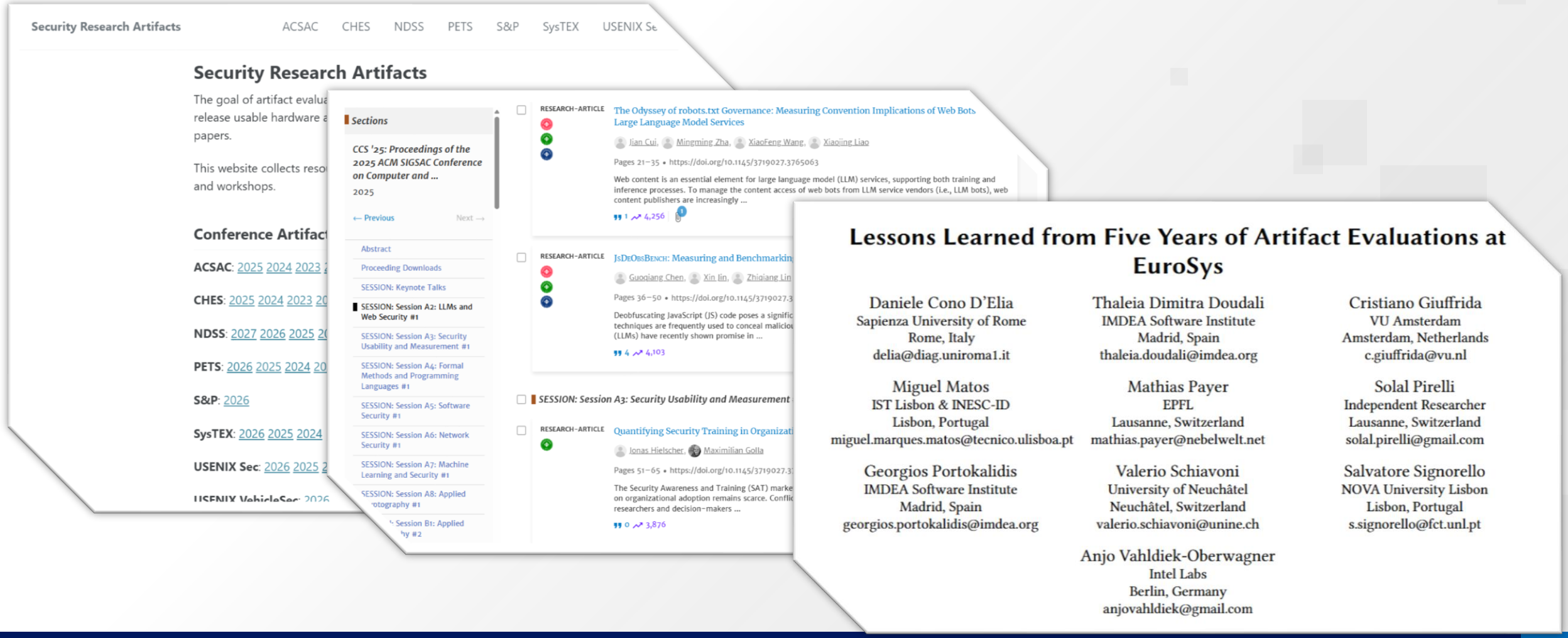
# ReproDB:
# An Open Platform for Discovering Research Artifacts and Analyzing Artifact Evaluation in Security and Systems

Anjo Vahldiek-Oberwagner (Intel),
Marton Bognar (DistriNet, KU Leuven),
Salvatore Signorello (NOVA University Lisbon)

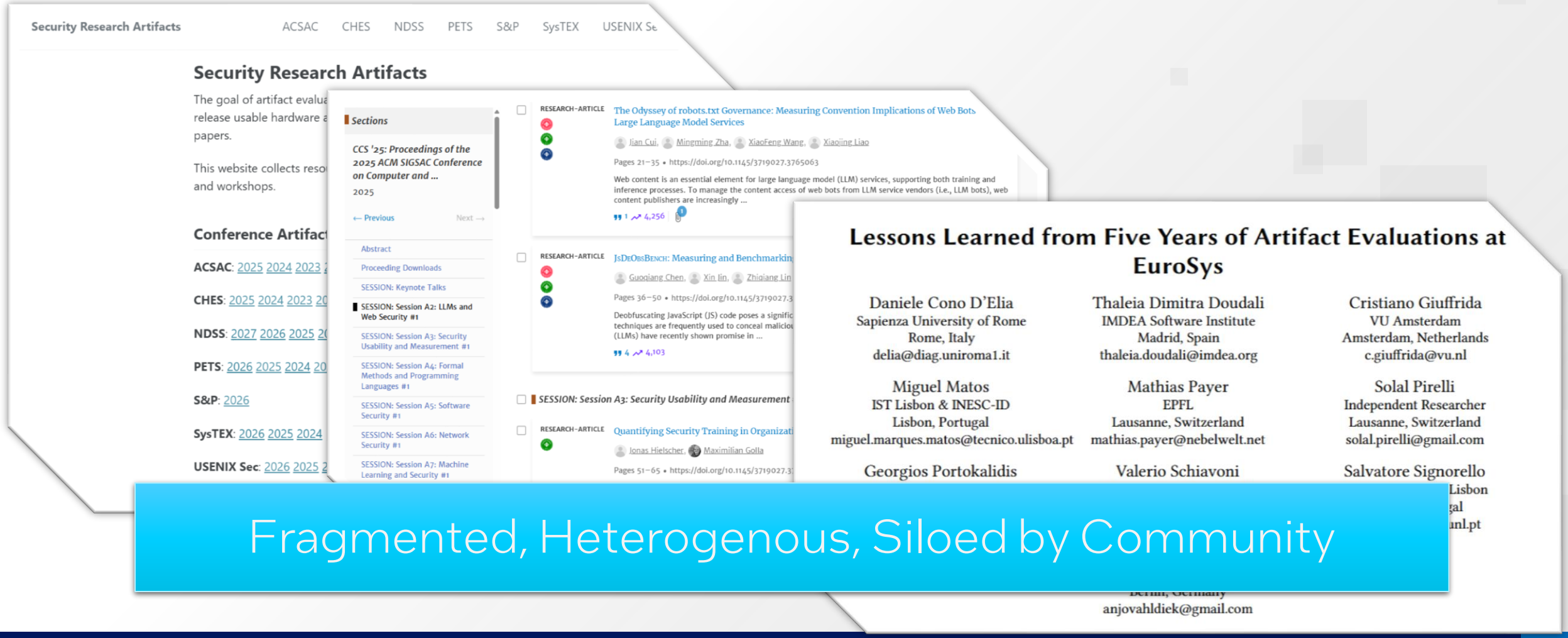
# Artifact Evaluation in Security & Systems



# Artifact Evaluation Documentation in Security and Systems



# Artifact Evaluation Documentation in Security and Systems



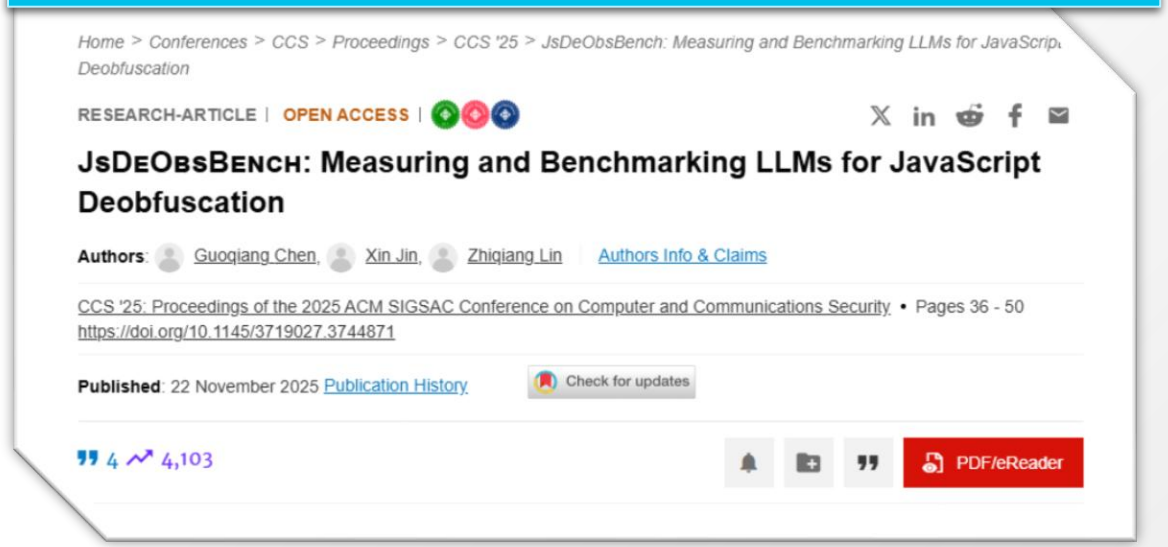

Fragmented, Heterogenous, Siloed by Community

# Finding Artifacts is Hard

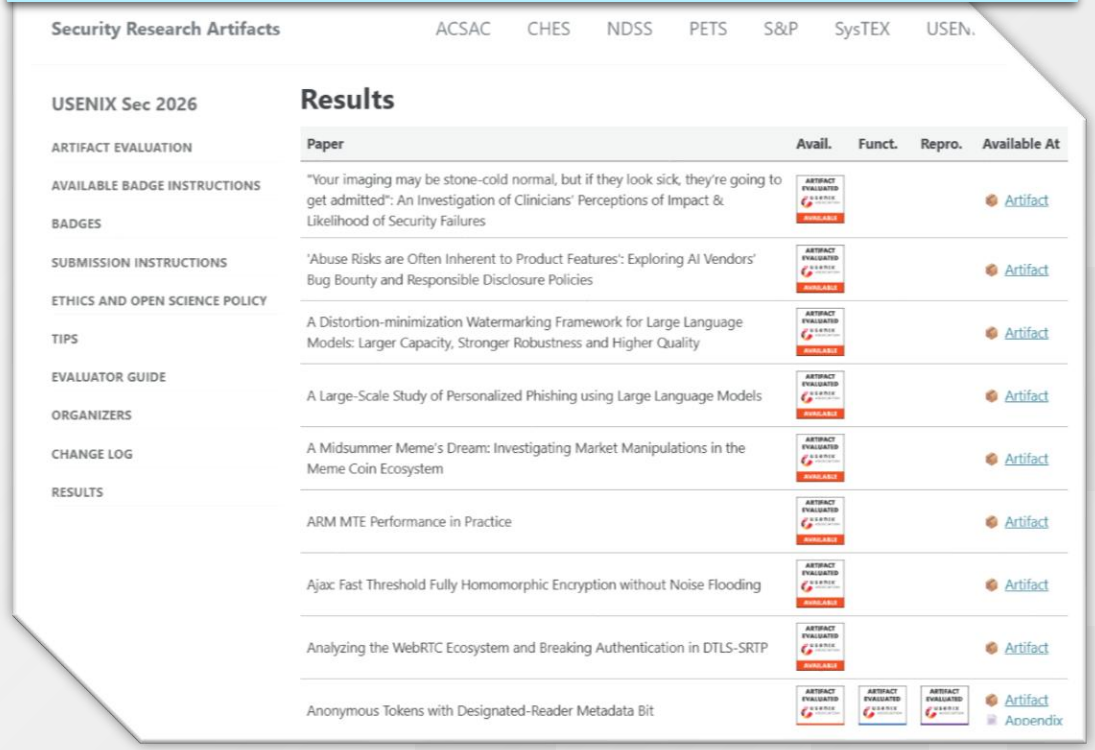
50-80% of papers include links to artifacts [1]

**Artifact and Leaderboard.** We have released our datasets and code in https://github.com/OSUSecLab/JSDeObsBench, and created a leaderboard in https://jsdeobf.github.io/.

ACM DL/Xplore don't associate artifacts to papers

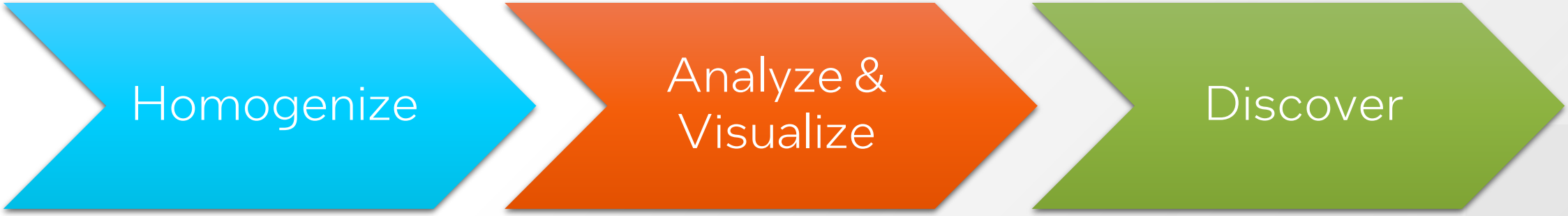



Sec/Sysartifacts host per Venue per Year lists




[1] Daan Vansteenhuyse, Arthur Bols, Lieven Desmet, Victor Le Pochat,Jo Van Bulck, Marton Bognar: "Not All Those Who Share Are Lost: Analyzing 25 Years of Cybersecurity Artifact Sharing Practices Through Automated Discovery", USENIX Security 2026

# Need for an Open Platform: ReproDB

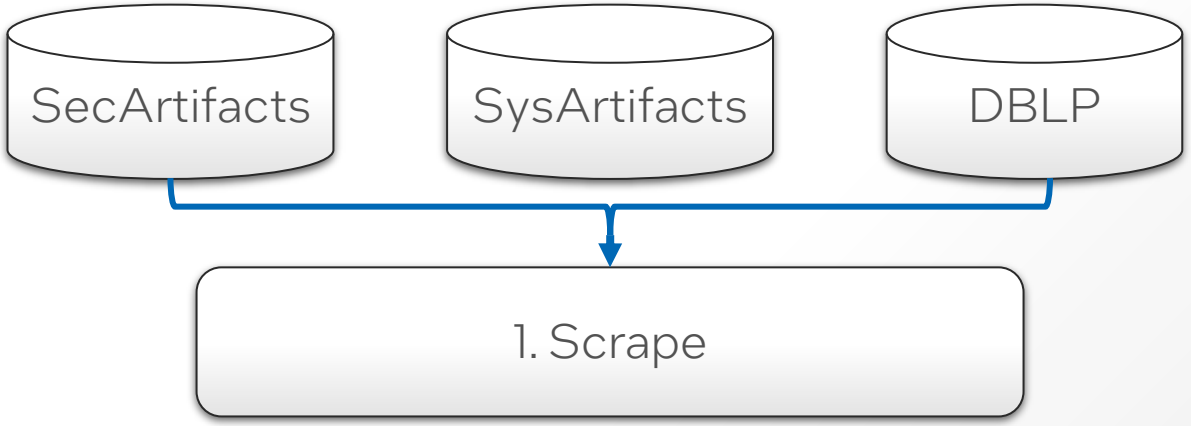



- Single Dataset
- Open-Source
- Continuous

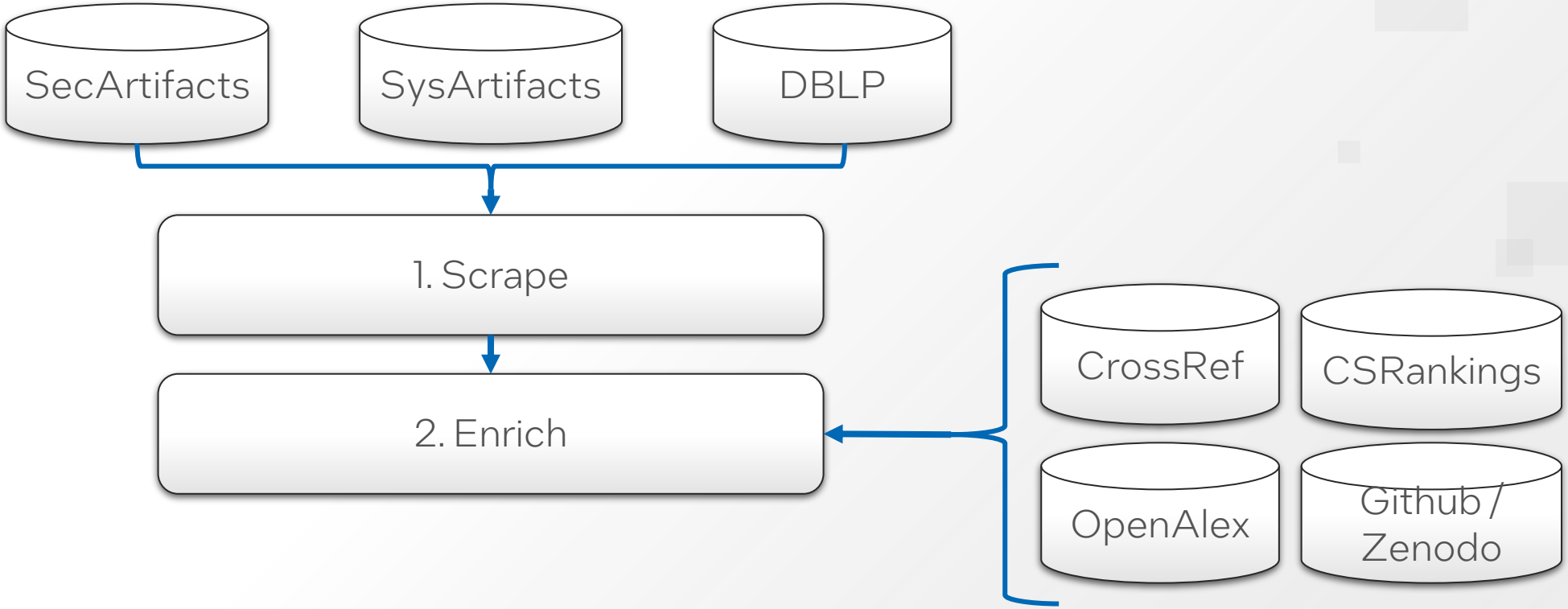
- Cross-Community
- Metrics
- Visualize Labor

- By topic, authors, …
- Accessible at large

# Scrape, Enrich, Analyze, Publish

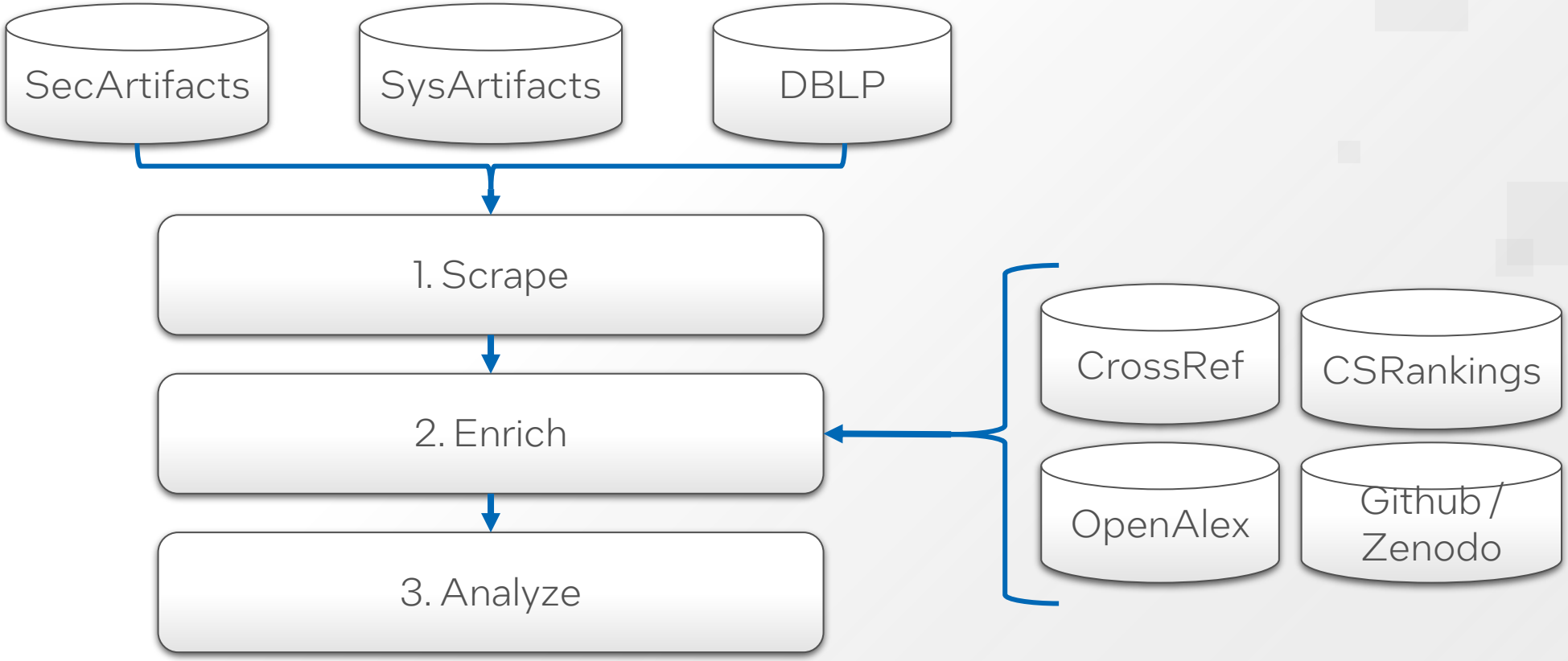


# Scrape, Enrich, Analyze, Publish

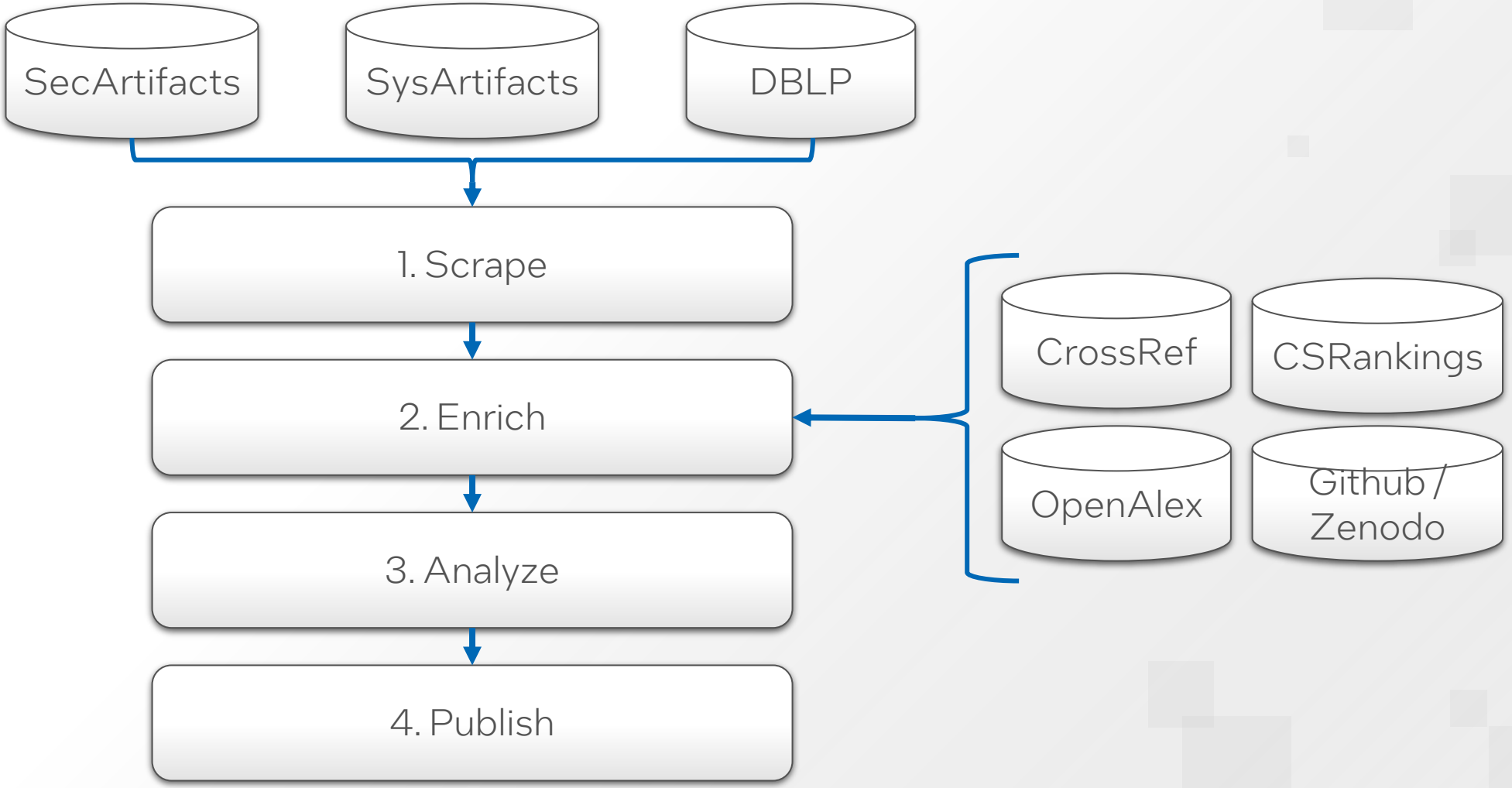



92% of artifacts are associated to authors,
57% of author affiliations are resolved.

# Scrape, Enrich, Analyze, Publish



# Scrape, Enrich, Analyze, Publish



# Metrics: Combined Author Score

Reflecting Artifact **Creation** and **Evaluation Labor** in a Single Metric

$$S(a) = \sum_{i \in Artifacts(a)} (A_i + F_i + R_i) + \sum_{j \in AEService(a)} (M_j \times 3 + C_j \times 2)$$

(1 point per badge)

(3 points per AE membership + 2 points as chair)

$$\text{A:E} = \frac{\text{Creation Score}}{\text{Evaluation Score}}$$

# Metrics: Artifact Rate & Reproduced Rate

Relative Metrics measuring breadth and depth of artifact creation

$$\mathrm{AR}(a) = \frac{\#\text{ of papers with badged artifacts}}{\text{total number of papers at AE conferences}}$$

$$\mathrm{RR}(a) = \frac{\#\text{ of papers with reproduced/reuseable badge}}{\text{total number of papers with any badge}}$$

# The Dataset:
# 13 Venues in Security and Systems (2017 – 2026)

2,831
Evaluated Artifacts

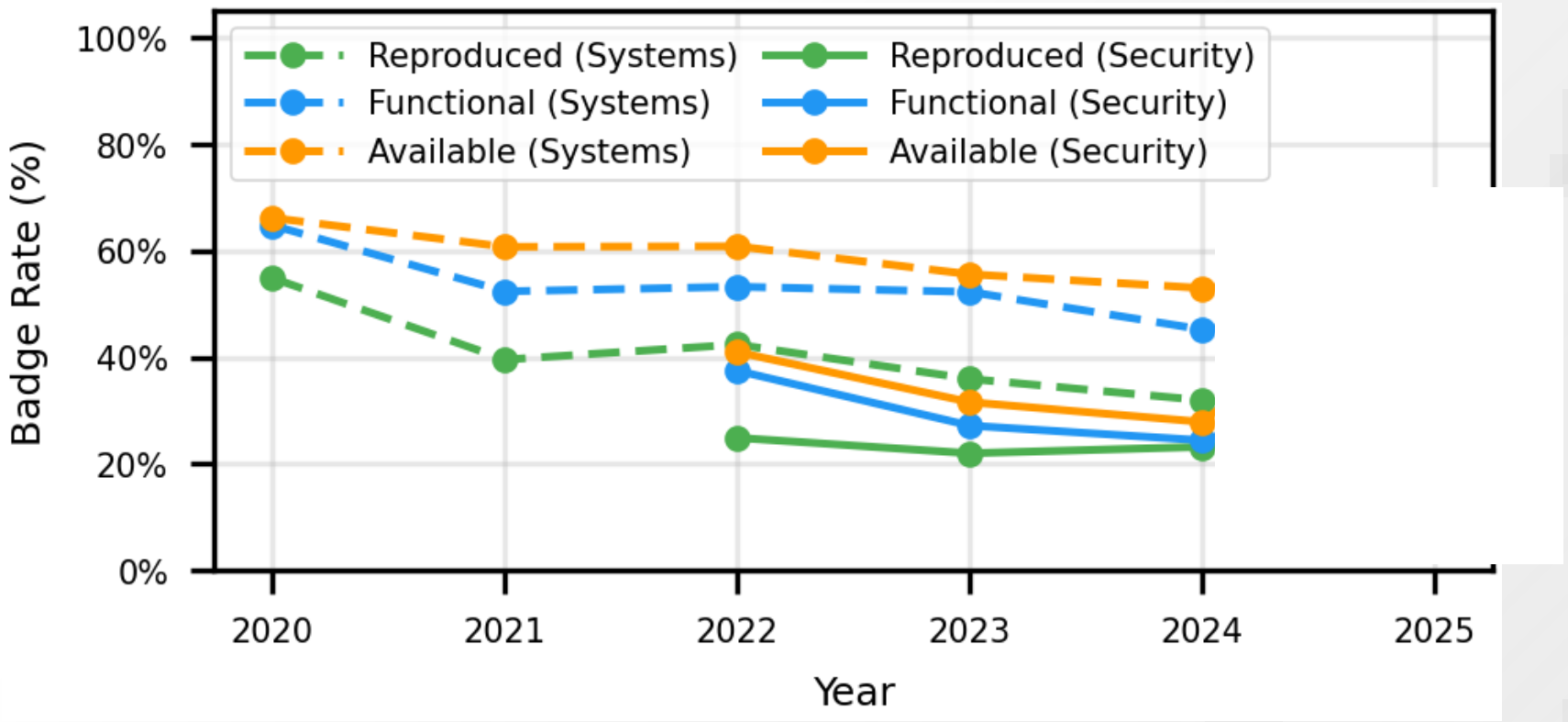
8,139
Artifact Authors

2,458
Artifact Evaluators

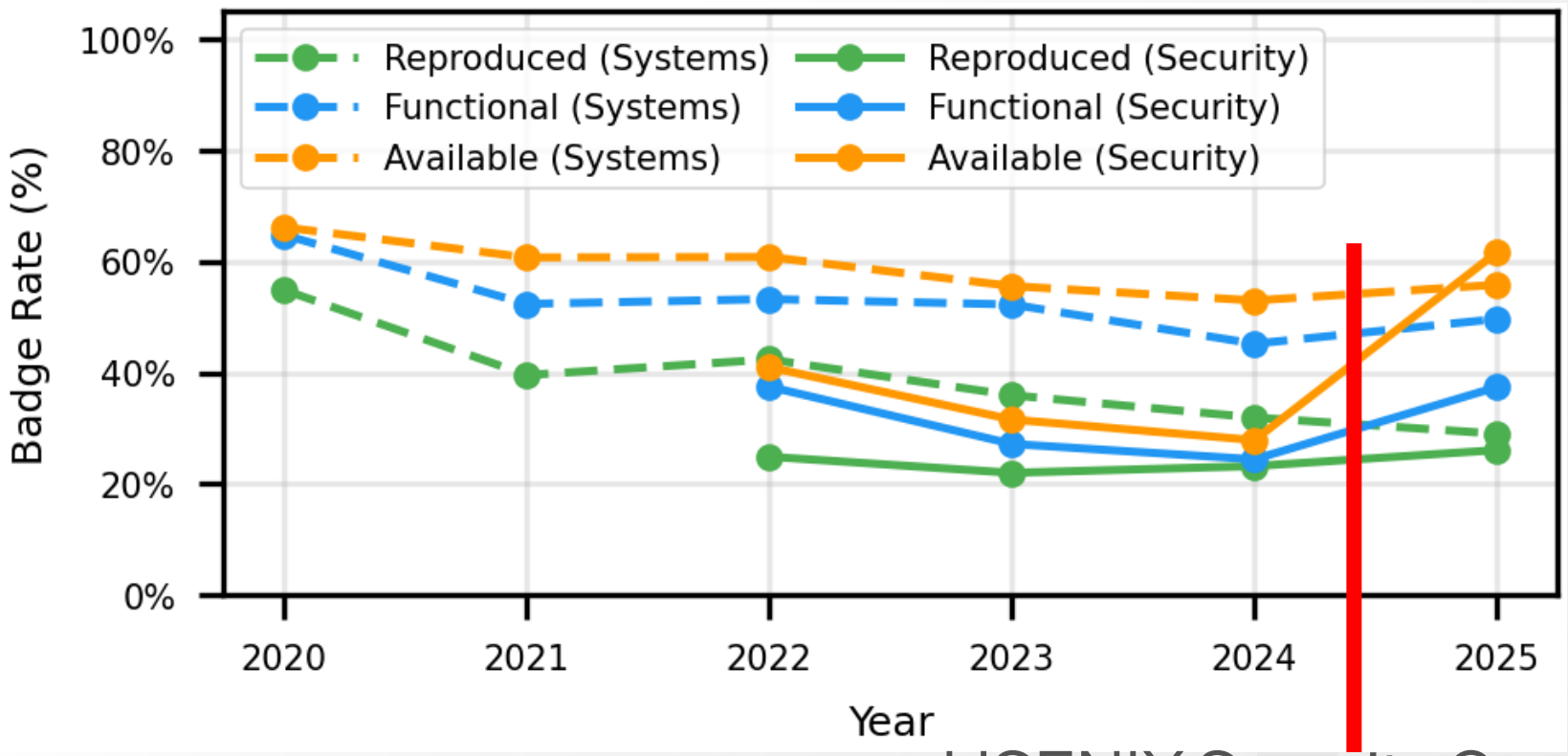
902
Institutions

| | Security | Systems | Overall |
|---|---|---|---|
| Artifacts | 2091 | 740 | 2831 |
| Evaluated | 393 (19%) | 0 | 393 (14%) |
| Available | 1345 (64%) | 722 (97%) | 2067 (73%) |
| Functional | 1189 (57%) | 642 (87%) | 1831 (65%) |
| Reproduced | 759 (36%) | 461 (62%) | 1220 (43%) |

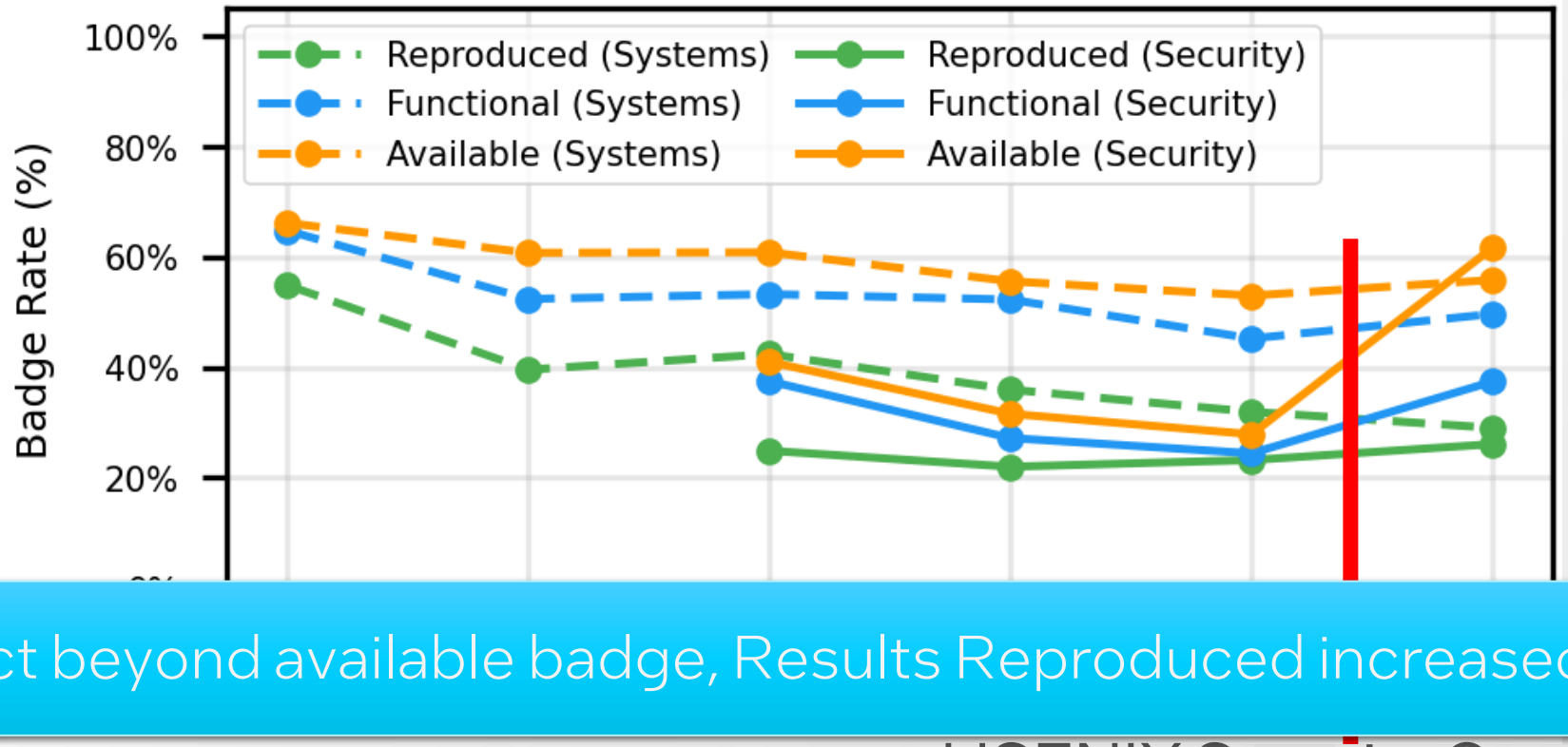
# Findings: How to increase AE participation?



# Findings: Largest Impact: Open-Science Policy @ USENIX Security



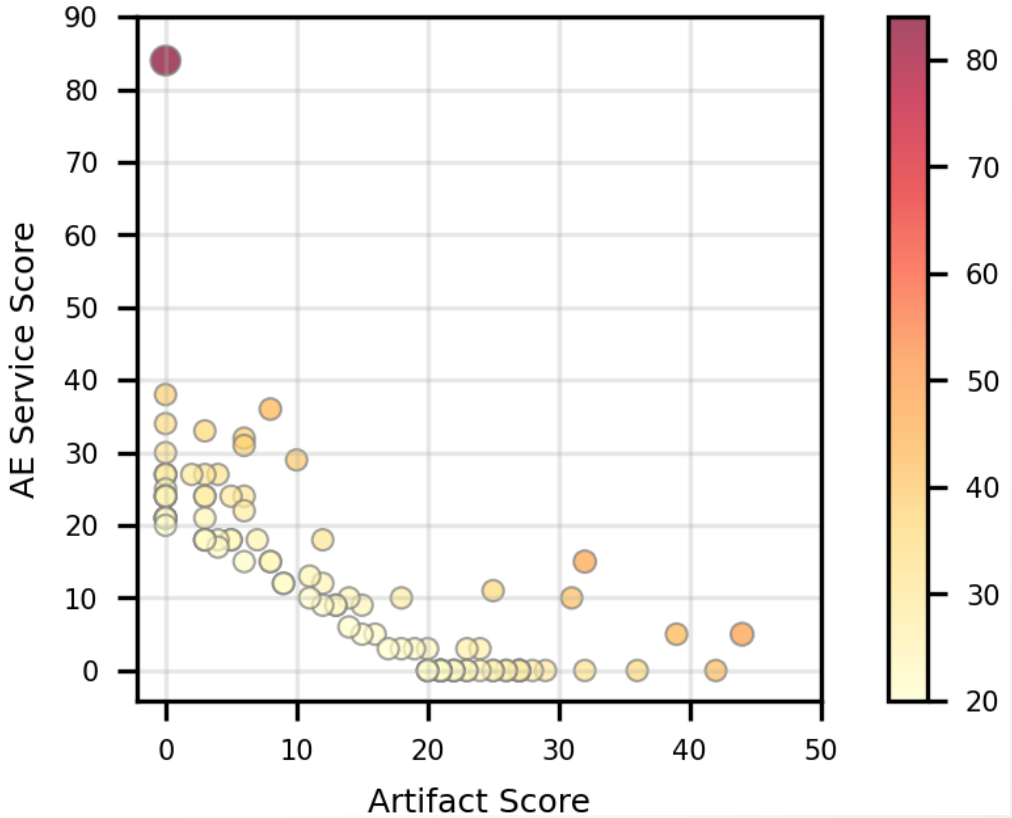
# Findings: Largest Impact: Open-Science Policy @ USENIX Security



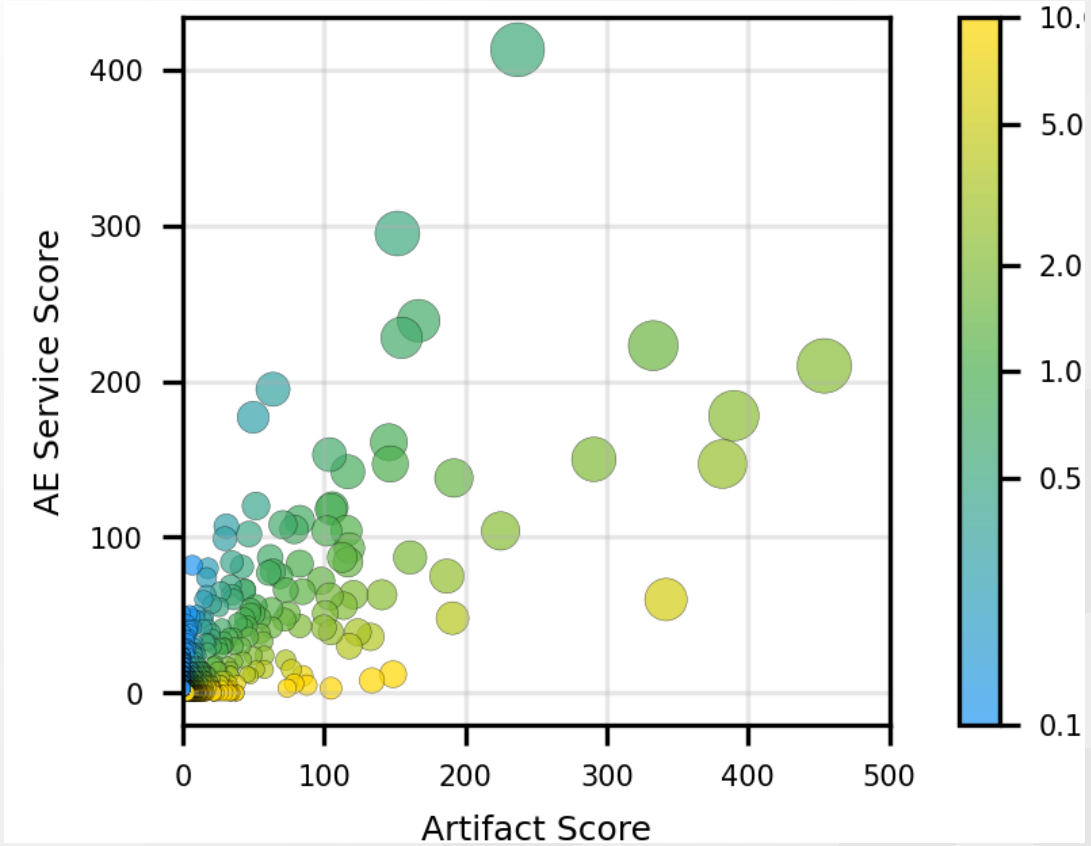

Effect beyond available badge, Results Reproduced increased by ~14%

# Findings: Ecosystem Roles

Authors

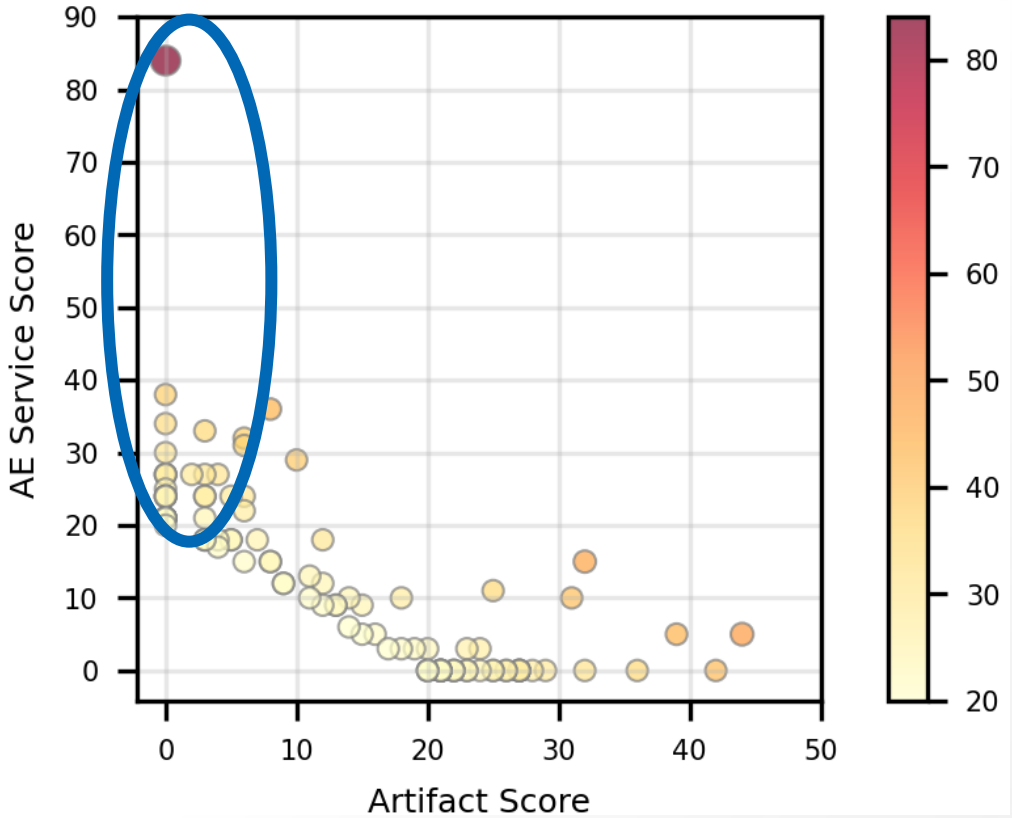



Institutions

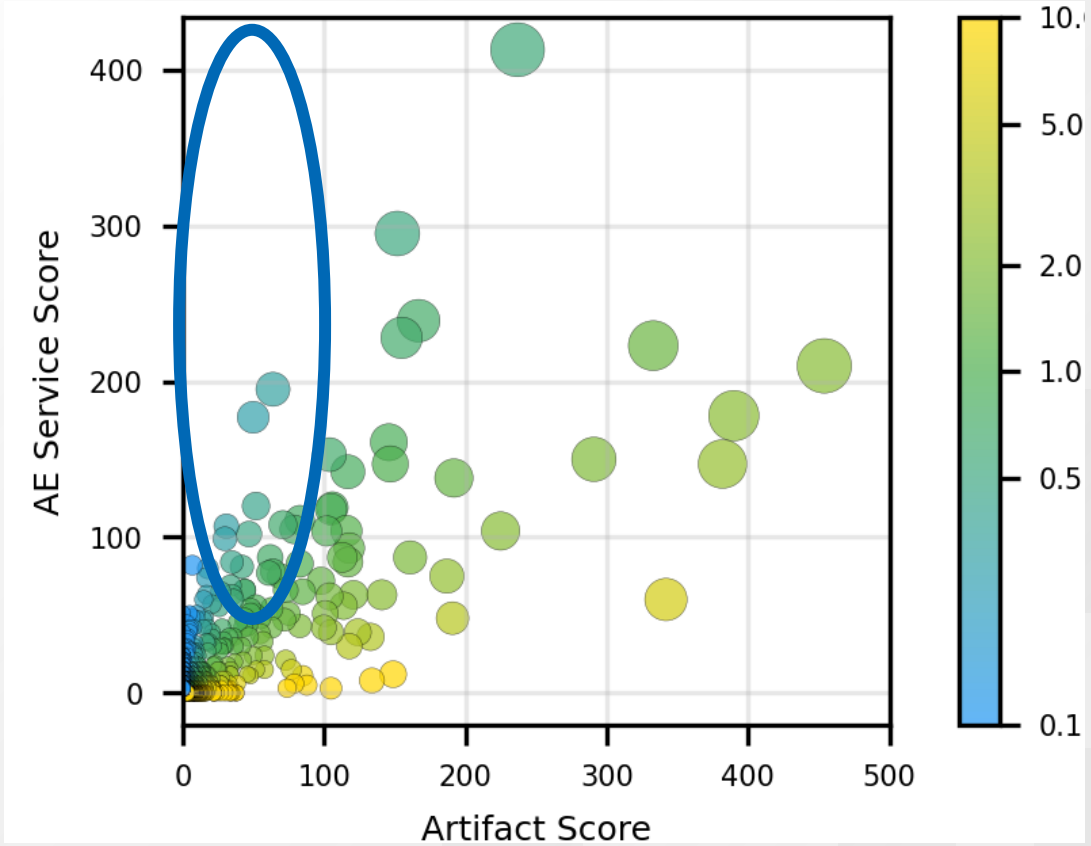


# Findings: Ecosystem Roles: Service

Authors

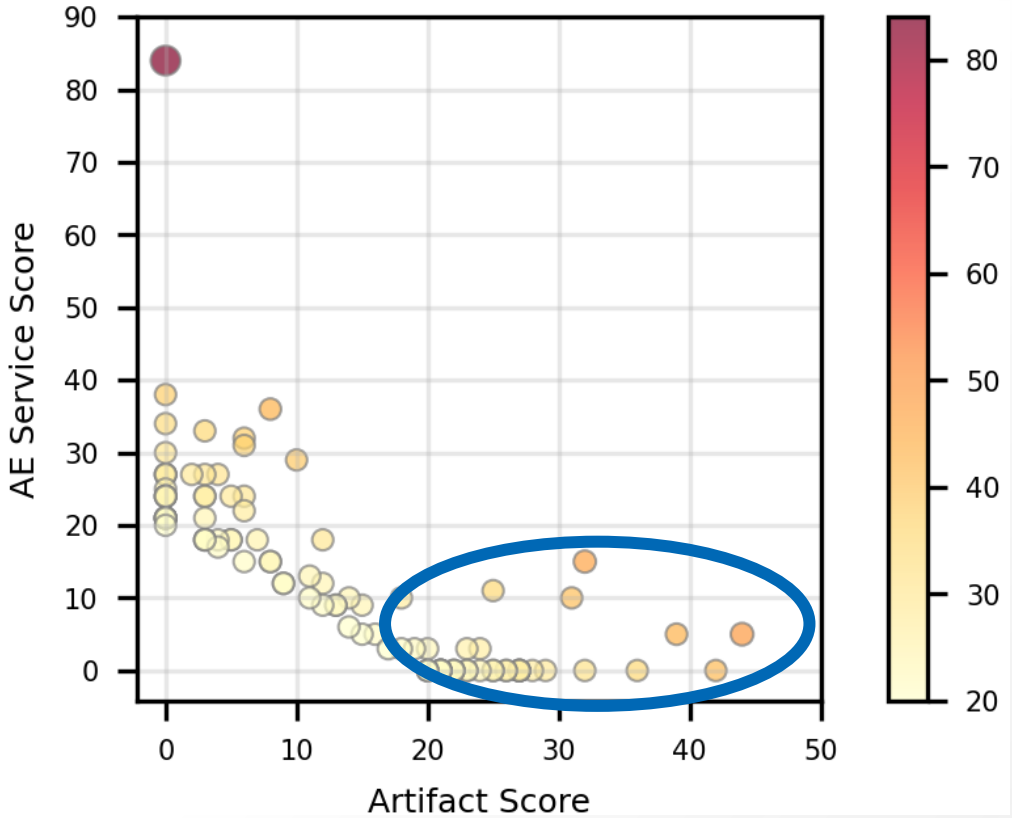



Institutions

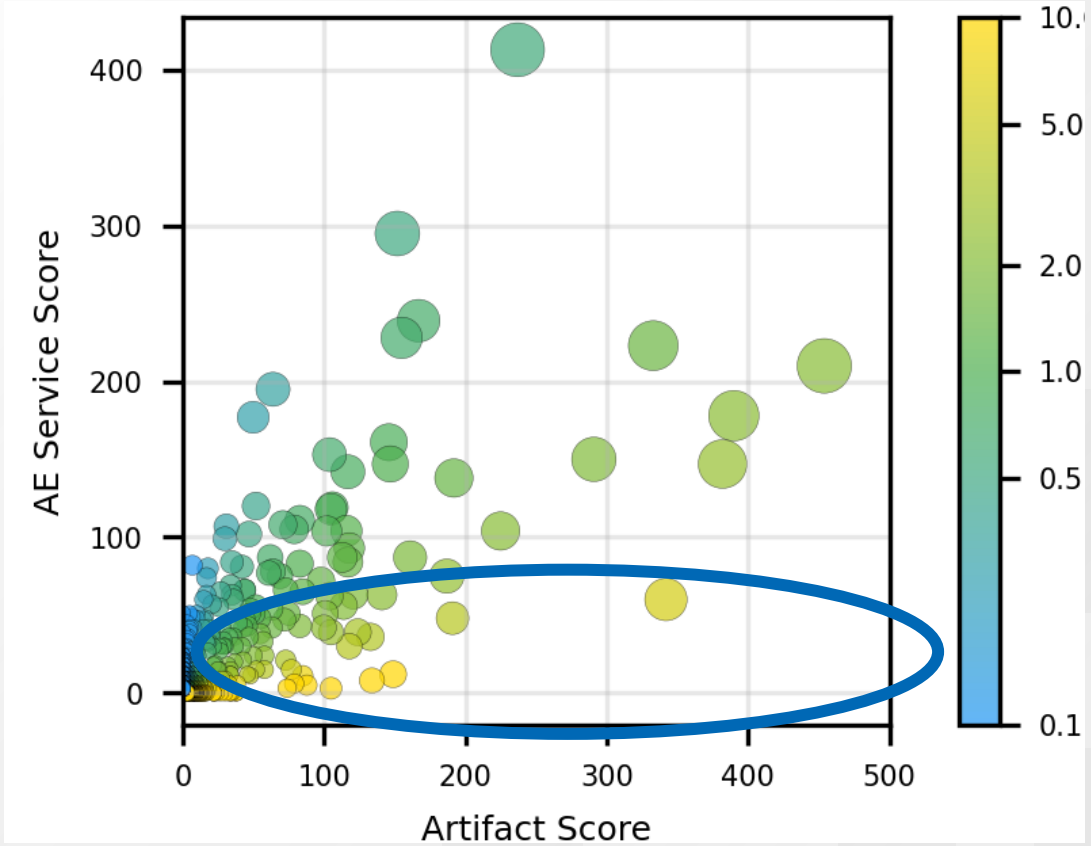


# Findings: Ecosystem Roles: Creators

Authors

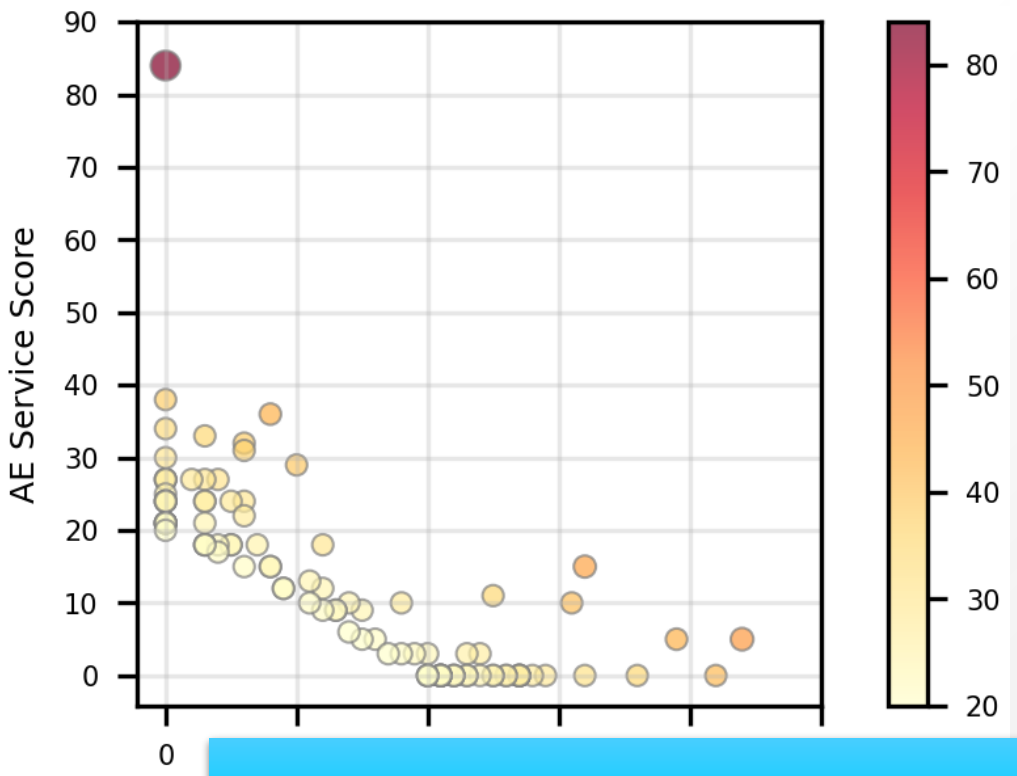



Institutions

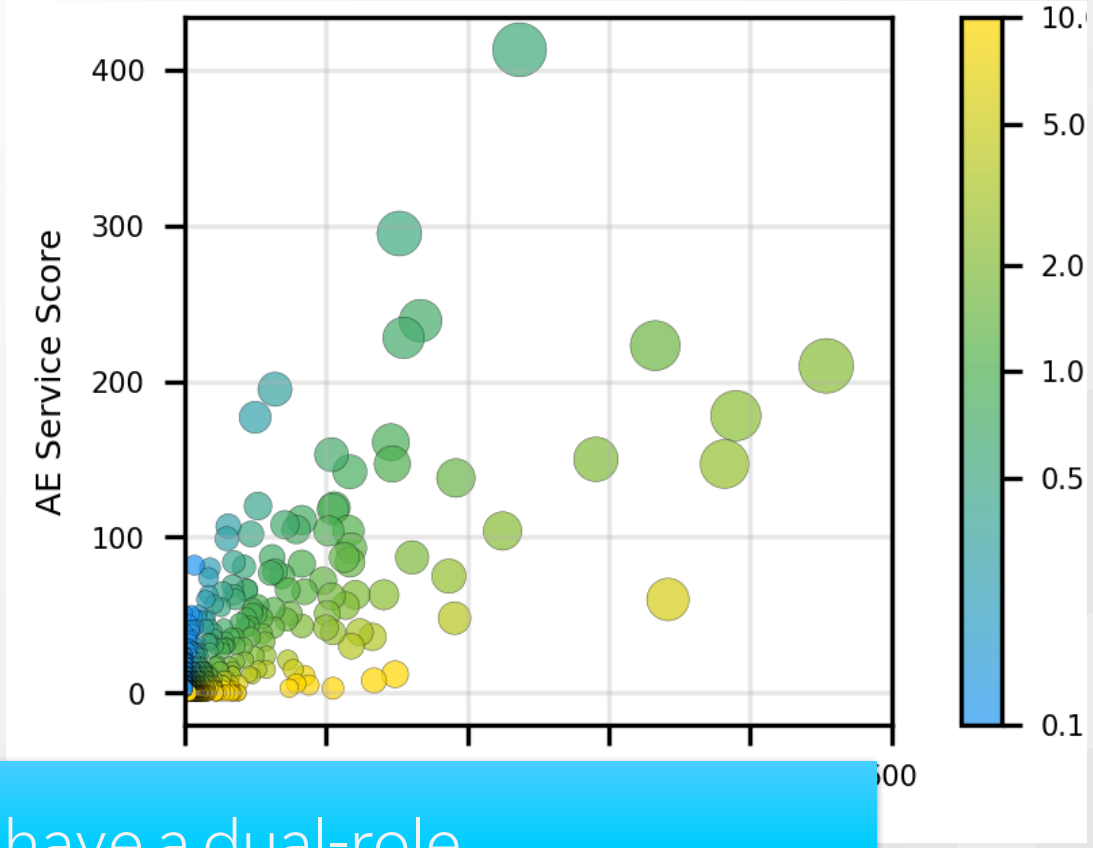


# Findings: Ecosystem Roles

Authors

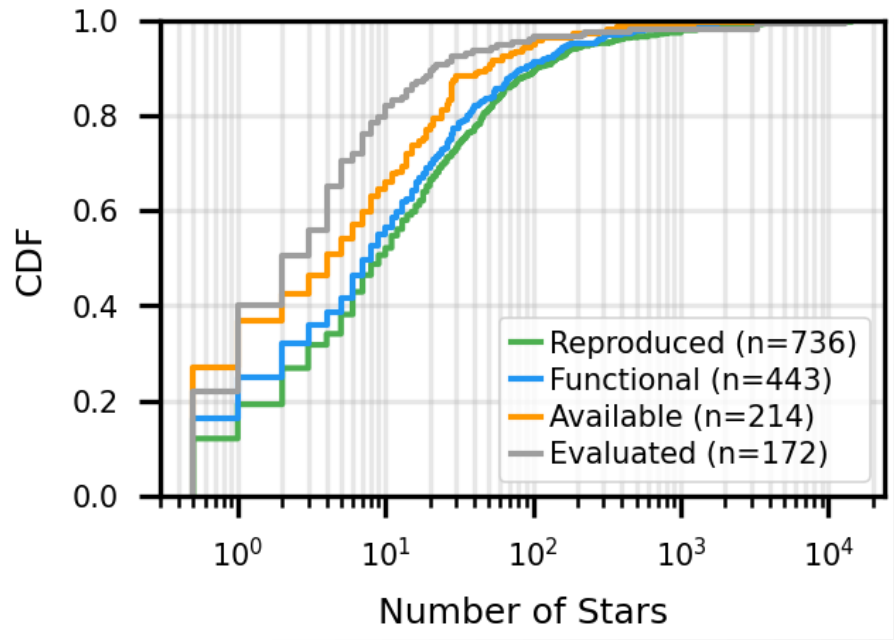


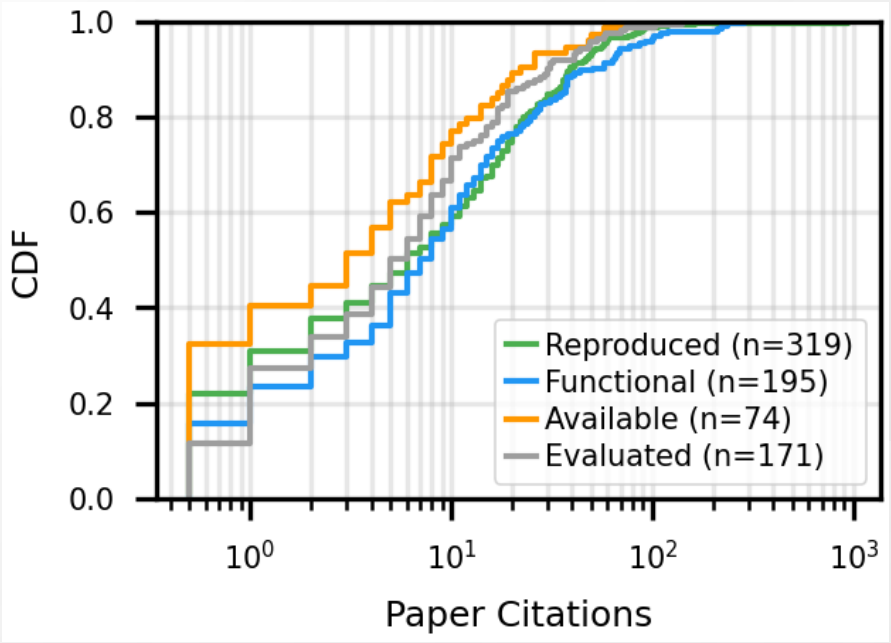

Institutions




50% of top authors have a dual-role.

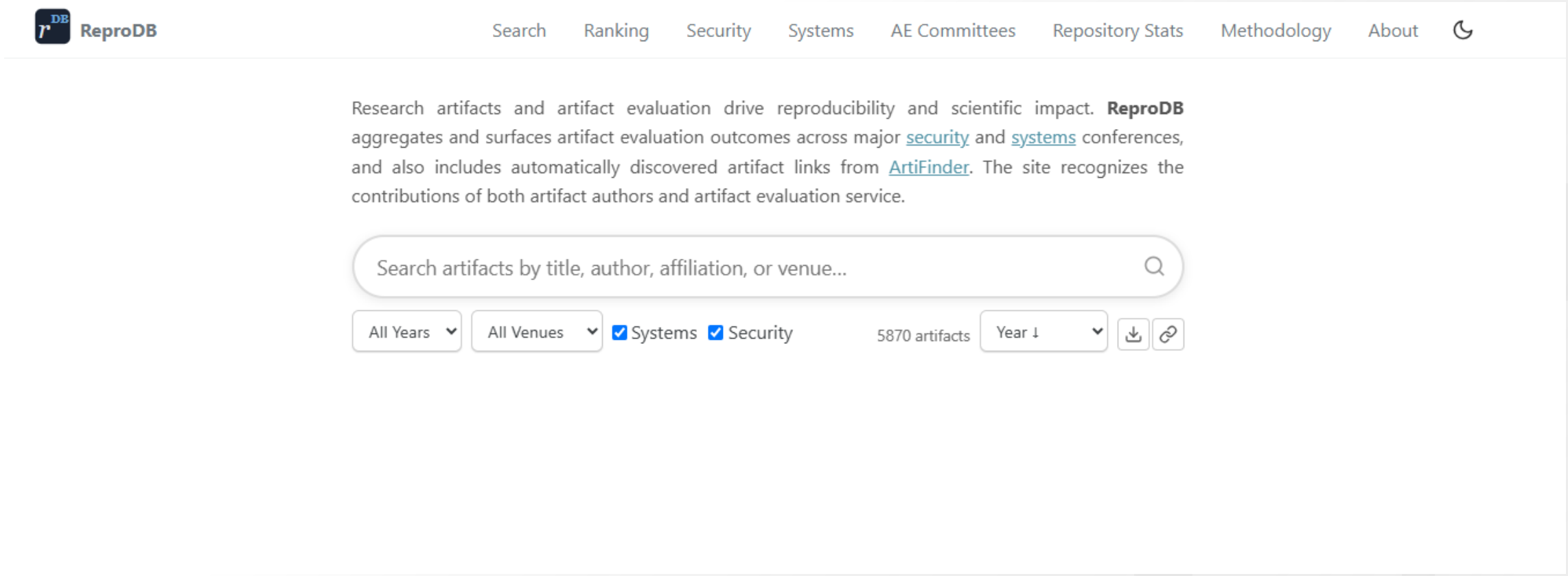
# Findings 3: Do badges promote papers?




In Systems, papers with reproduced artifact earn 2.2x median Github stars, and 1.8x median citations compared to available-only.

# Demo of Search



# Summary: ReproDB




- Infrastructure to enable artifact reuse and research
- Continuous measurement and analysis

## Available at reprodb.github.io

Contributions welcome: github.com/ReproDB

# Available at reprodb.github.io

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